

Reproducibility Guide — EMS Readiness Optimization

Why Reproducibility Matters

The EMS simulation uses stochastic processes (NHPP arrivals, random service times, dispatch tie-breaking). Without careful seed management:

- Results change between runs, making debugging impossible.
- Published results can't be verified.
- A/B comparisons of policies are confounded by randomness.

Architecture

SeedManager (`src/ems_readiness/utils/reproducibility.py`)

Central class that provides **deterministic, isolated** random number generators.

```
Master Seed (42)
├─ hash("arrivals")   → RNG for arrival generation
├─ hash("service")    → RNG for service times
├─ hash("dispatch")   → RNG for dispatch tie-breaking
└─ hash("simulation") → RNG for SimPy engine
```

Key properties:

- **Deterministic**: Same master seed → same derived seeds → same results.
- **Isolated**: Changing the arrival stream doesn't affect service-time draws.
- **Cached**: Requesting the same component twice returns the same RNG instance.

Configuration

```
# configs/reproducibility.yaml
reproducibility:
  master_seed: 42
  components:
    - arrivals
    - service
    - dispatch
    - simulation
```

Override via environment variable:

```
export EMS_MASTER_SEED=12345
python run_simulation.py
```

Usage

Basic

```
from ems_readiness.utils.reproducibility import SeedManager

sm = SeedManager(master_seed=42)
rng = sm.get_rng("arrivals")
values = rng.random(10) # always the same 10 floats
```

From Config File

```
sm = SeedManager.from_config("configs/reproducibility.yaml")
```

Batch Replications

For N replications with different but reproducible seeds:

```
for rep in range(10):
    sm = SeedManager(master_seed=42 + rep)
    engine = SimulationEngine(seed=sm.get_seed("simulation"))
    engine.run()
```

This matches the existing `BatchRunner` pattern (`seed_base + rep`).

Recording Metadata

```
meta = sm.get_metadata()
# {'master_seed': 42, 'components': {'arrivals': 17239847, ...}}
import json
with open("results/run_metadata.json", "w") as f:
    json.dump(meta, f, indent=2)
```

Reproducing a Past Experiment

1. Find the `master_seed` from the run's metadata or config.
2. Set it in `configs/reproducibility.yaml` or `EMS_MASTER_SEED`.
3. Run the same code version (check the git commit hash).
4. Results will be bit-for-bit identical.

Integration with Existing Code

The simulation engine (`engine.py`) already accepts a `seed` parameter:

```
engine = SimulationEngine(
    config=config,
    allocation=allocation,
    demand_data=demand_data,
    seed=sm.get_seed("simulation")
)
```

The `BatchRunner` (`runner.py`) already uses `seed_base + rep` .

To integrate `SeedManager`, replace the literal seed base:

```
sm = SeedManager(master_seed=config.get("master_seed", 42))
for rep in range(n_replications):
    rep_sm = SeedManager(master_seed=sm.get_seed("simulation") + rep)
    # ...
```

Best Practices

1. **Always record the master seed** alongside results.
2. **Never share RNGs** between independent components.
3. **Pin the git commit** — code changes can alter call order even with same seed.
4. **Use the environment variable** for CI/CD sweeps over seeds.
5. **Run ≥ 30 replications** for publishable statistics; report the seed range.